

LABWORK 6

This lab aims to help you practice pointers, arrays, and functions in C by solving real-life inspired problems. You will implement 10 functions, each focusing on different array manipulations and pointer usage.

Function 1: max_temperature (5 pts)

- **Return Type:** `int`
 - **Parameters:** `const int A[], int size`
 - **Description:** Find and return the maximum temperature of the week.
 - **Example:**
 - Input: {23, 25, 28, 27, 22}
 - Output: 28
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Function 2: min_stock_level (5 pts)

- **Return Type:** `int`
 - **Parameters:** `const int A[], int size`
 - **Description:** Find the minimum stock level in inventory.
 - **Example:**
 - Input: {10, 5, 8, 12}
 - Output: 5
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Function 3: reverse_array (5 pts)

- **Return Type:** `void`
 - **Parameters:** `int A[], int size`
 - **Description:** Reverse an array of sales data in place.
 - **Example:**
 - Input: {100, 200, 300}
 - Output: {300, 200, 100}
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Function 4: count_even_odd_days (10 pts)

- **Return Type:** `void`
 - **Parameters:** `const int A[], int size`
 - **Description:** Count how many days had even vs odd temperature readings.
 - **Example:**
 - Input: {22, 23, 24, 25}
 - Output: Even: 2, Odd: 2
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Function 5: sum_positive_sales (10 pts)

- **Return Type:** `int`
 - **Parameters:** `const int A[], int size`
 - **Description:** Calculate the total positive sales. Negative numbers indicate returns.
 - **Example:**
 - Input: {100, -20, 50, -10}
 - Output: 150
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Function 6: multiply_daily_steps (10 pts)

Return Type: void

- **Parameters:** `int A[], int size, int factor`
 - **Description:** Multiply all daily step counts by a given factor (e.g., step goal adjustment).
 - **Example:**
 - Input: {1000, 2000, 1500}, factor=2
 - Output: {2000, 4000, 3000}
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Function 7: pointer_increment_inventory (10 pts)

- **Return Type:** `void`
 - **Parameters:** `int *A, int size`
 - **Description:** Increment each inventory item by 1 using pointer arithmetic.
 - **Example:**
 - Input: {5, 10, 15}
 - Output: {6, 11, 16}
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Function 8: find_max_sales_pointer (10 pts)

- **Return Type:** `int*`
 - **Parameters:** `int A[], int size`
 - **Description:** Return a pointer to the day with maximum sales.
 - **Example:**
 - Input: {200, 450, 300}
 - Output: Pointer to 450
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Function 9: count_between_values (10 pts)

- **Return Type:** `int`
 - **Parameters:** `int A[], int size, int low, int high`
 - **Description:** Count the number of days where sales are between low and high inclusive.
 - **Example:**
 - Input: {100, 200, 300, 400}, low=150, high=350
 - Output: 2
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Function 10: swap_days (10 pts)

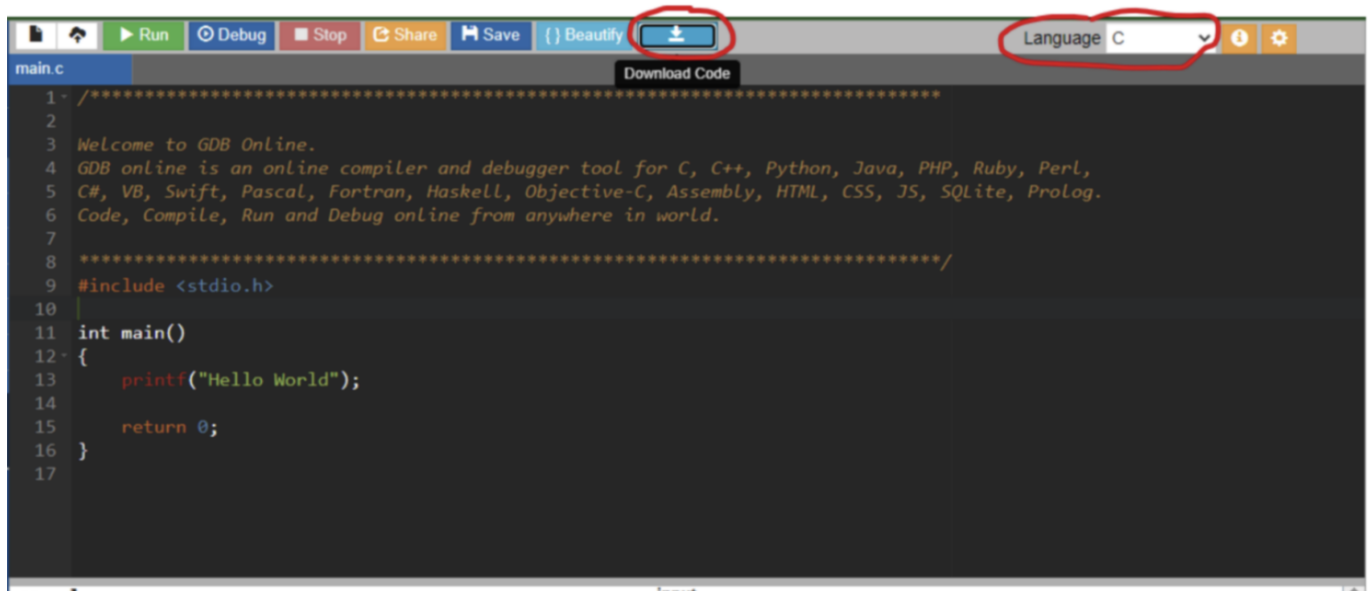
- **Return Type:** `void`
 - **Parameters:** `int *a, int *b`
 - **Description:** Swap sales of two days using pointers.
 - **Example:**
 - Input: a=100, b=300
 - Output: a=300, b=100
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Main Function (15 pts)

- **Function Name:** `main`
 - **Return Type:** `int`
 - **Parameters:** `none`
 - **Description:** Ask the user for input for temperatures, inventory, sales, steps, and arrays for each function.
 - **Example:**
 - Input: {20, 22, 24, 26, 26, 24, 22, 20}
 - Output: 1
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Submission Format

- Submit your `.c` file named as:
`name_surname.c`



Important Notes

- Late submissions will not be accepted.
- Follow function naming and signature exactly.
- Modularize your code using functions.
- Test each function with real-life examples.
- Use loops, pointers, and array arithmetic; do not use extra libraries beyond `stdio.h` and `string.h`.